Trading environments

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Many organisations are now adopting the Internet as a new channel to market — and in particular are publishing their product catalogues on-line with the capability of accepting orders. However, as more catalogues emerge the reliance upon discrete catalogues will become less tenable for suppliers and buyers alike — buyers would be unlikely to find the best merchant without some sort of sophisticated help, and similarly suppliers would find it hard to reach their potential customer base. This approach, driven by the supplier, will almost certainly be complemented by other kinds of market-place driven more from the buyer's perspective. Three main kinds of market-place are envisaged, namely seller-driven, buyer-driven and open markets.

In order to support these trading models, different approaches to system architectures are emerging, and these are also discussed.

1. Introduction

The recent growth in adoption of the Internet has provided a ubiquitous medium over which buyers and sellers can find one another, interact and ultimately transact — in other words an environment in which they can trade. All businesses need to both buy and sell in order to exist, while individual consumers will become increasingly receptive to an opportunity to buy more efficiently. The growth of the Internet will continue to percolate into the home, with new games consoles, the advent of digital interactive TV, more sophisticated screen-based telephone devices, and personal digital assistants (PDAs). Electronic trading, therefore, has the potential to affect the very heart of businesses and consumers alike — provided, of course, that it is able to offer real benefits over traditional means of trading. The benefits that will lead to widespread adoption of electronic trading will only be realised through the development of innovative applications, which will themselves become the basis for new electronic market-places. One key attraction of electronic trading is the prospect of a business tailoring its interaction with a customer based on its accumulated knowledge of that customer (often referred to as profiling). For instance, this could lead to a customer receiving an individual welcome, their own selection of items and information, receiving targeted advertising, and special offers.

In the real world there are a number of different types of market-place, for example:

- the retail shop,
- auctions,
- newspaper classified advertisements,
- stock or commodity markets.

This paper will discuss how different types of market-place are also anticipated to emerge in the on-line world, and identify some of the ways in which it is anticipated that electronic market-places will develop to become the trading environments of choice for businesses and consumers alike. Also considered will be the different architectures that are emerging to support environments that provide a focus for both buyers and sellers.

2. Trading models

We are all familiar with the growing number of businesses that are successfully exploiting the Internet as a new channel to market. Examples of the many businesses now doing a significant proportion of their business over the Internet include Barnes & Noble [1] and Dell Computer [2]. Many other companies, BT included, recognise the importance of the Internet as a channel and are putting in place their own product catalogues (e.g. the BT Shop [3]). There are also examples of start-up businesses that are using the Internet as their only channel to market — the best known of these being Amazon — the on-line bookstore.

In general, the companies experiencing the greatest success with on-line catalogues are those who already have a strong brand (or start-ups who have invested heavily in creating one). Among small-to-medium sized enterprises (SMEs) there have also been examples of significant new business gained through the Internet channel, particularly where the business serves a niche market. One good example of this is Trophy Miniatures, a maker of pewter-rich model soldiers who have gained a significant growth in business since promoting themselves on the Web, as part of the South Wales virtual business community trials being carried out by the University of Cardiff [4, 5].

However, not all attempts to establish an Internet channel have been so successful as the examples above. As more businesses compete for a share of this channel, then it will be important for businesses to understand the market dynamics in order to succeed.

There are a number of ways in which market-places can be characterised, the most significant of which are:

- the driver for the market-place (is it supplier-driven, buyer-driven or neither?),
- what is being promoted (products, requirements or both?),
- type of participant (business-to-business, business-to-consumer or consumer-to-consumer?)[1],
- importance of time-lines (real-time, near real-time or asynchronous?).

It is particularly instructive to construct a model from the first two of these characteristics. This view has two dimensions — whose brand attracts participants and what is advertised (offers from sellers, or requirements (bids) from buyers?). In this model:

- a seller-driven market is deemed to be one where a supplier with a strong brand can attract buyers to their own market-place (generally based on a catalogue),
- a buyer-driven market is one where one or more strong buyers can attract their suppliers to participate in a market-place for the exclusive benefit of those buyers,
- an offer-led market is deemed to be one where it is the seller that advertises their products or services,
- a requirement-led market is one where it is the buyer that advertises their requirements (a requirement is analogous to a bid in stock-broking terminology).

In addition, there are open markets in which it is neither a strong buyer nor a strong seller that attracts participants, but the brand of a strong market-operator, and markets where either the buyer or seller may advertise, depending on the particular circumstances prevailing. This model is illustrated in Fig 1, which also describes the kinds of market that would typically exist at each classification.

The model derived in Fig 1 can be specialised depending on the other relevant characteristics of the market-place — such as whether the supplier is primarily focused on business or consumer customers. In addition, it can be helpful to position types of market on this model — for instance, it becomes self-evident that consumer-to-

[1] Some people distinguish administration-to-business from business-to-business where administration here means any government department, agency, or similar. The trading process, needs and requirements of such can be somewhat different from those of business.

drive - whose brand attracts participants?

lead - what is advertised?	seller	market-place	buyer	
offer	supplier with strong brand in their market, publishing a catalogue direct banking	suppliers without very strong brand, publishing product catalogues in a strong market-place auctions	Hub and Spoke model focused on very strong buyer(s), incorporating product catalogues	offer-led markets
either		classified adverts, spot markets	Hub and Spoke model incorporating tenders as well as product database	jointly-led markets
requirement	supplier with strong brand selling customised products	service allowing buyers to issue RFQs for products and services	very strong buyer issuing ITTs	requirement-led markets
	seller-driven markets	open markets	buyer-driven markets	

Fig 1 Markets may be described by the dimensions ‘lead’ and ‘drive’.

consumer market-places will inevitably be types of open market since neither the buyer nor seller will have a strong brand in their own right.

Just as in the real world, the majority of today's trading applications are seller-led. Any supplier wishing to reach out into the on-line market-place can readily set up an independent on-line store to market his products or services. However, the new on-line medium is also introducing new ways to find information, suppliers and product advice. This is causing a fundamental shift in the relationship between buyers and sellers, with the real potential to empower the buyer rather more than has been hitherto possible. These changes will manifest themselves in a number of ways, one of which is likely to be the growth in open-markets operated by intermediaries (perhaps particularly for commodities with complex requirements/specifications such as insurance and holidays). Finally, there is an emerging opportunity to provide procurement solutions for major corporations who are motivated by the need to control their total purchasing expenditure and rationalise their supplier base. This new application will primarily address maintenance, repair and operations (MRO) spend and be tightly integrated into the mainstream enterprise information systems such as SAP or BAAN, etc. The following sections examine the trends and developments in each of the three vertical columns in Fig 1.

2.1 Seller-driven regime

At the present time, most companies setting up to trade on-line and service their customers through this new channel to market do so unilaterally. This approach has been adopted successfully by the examples cited above [1—5]. In almost every case the suppliers will be advertising their products or services in some kind of catalogue. Consequently these applications will mostly belong on the seller-driven, offer-led segment of the market-space. One of the major differentiators in this space will be whether the application is to be focused on a business customer, a consumer or, as can often be the case, both business and consumer.

Business customers will only use the on-line medium if it offers significant advantages over traditional means. They will be motivated to be able to transact quickly and efficiently, and may well be put off by complex sites offering a 'shopping experience'. Moreover, business-to-business trading models will become more standardised — automating the interface between the buying and selling organisations. This will inevitably mean that trading environments aimed at business-to-business markets will need to adopt standards as and when they reach a critical mass of adoption. Suppliers will generally be focusing on providing service to their existing accounts, and will need to be able to reflect customer-specific pricing and service to individual buyers. Moreover, as buyer-driven MRO procurement systems are deployed, then suppliers may need to be able to establish a presence on a number of their customers' MRO systems as well as maintaining their open Internet presence. How this can be achieved will depend on the architecture of the procurement system, and is discussed in more detail below.

On the other hand, business-to-consumer applications will be competing for customers in a number of new and imaginative ways. It is far more likely that consumer-focused businesses will commission bespoke developments in order to differentiate themselves and incorporate new capabilities ahead of their competitors.

One of the crucial factors affecting the success of a trading application, especially for SMEs without a really strong brand, will be how effectively new buyers are attracted to the site. Today a supplier will submit details of their site to the different search engines, and may commission advertising banners and directory listings at Web portals and other strategic Web sites. However, the search engine approach is somewhat 'hit and miss' from a buyer's perspective. There is a relatively low likelihood of a search engine returning a meaningful result to a query because the HTML that makes up a Web page provides only rudimentary support for encoding information about the meaning of a Web page — through the use of meta-tags. Moreover, search engines are often unable to deal effectively with dynamic data-driven Web sites, which increasingly form the basis of on-line catalogues.

The acquisition of ViaWeb by Yahoo (now Yahoo Store) is also an interesting development. It is likely that Yahoo will be able to link their search engine to the Yahoo Store database. This will provide a double-edged tool — efficient cross-catalogue searches for buyers accessing the Yahoo portal and automatic inclusion in a well-known search engine for Yahoo Store merchants.

An emerging solution to this problem is the advent of XML (extensible markup language) discussed in more detail below. Once widely adopted, this will allow search-engines to 'understand' the nature of the content of a catalogue rather more effectively than at present and therefore allow buyers to locate products (and suppliers) rather more easily than at present. Once search engines are able to provide more meaningful product searches they will start to become the 'front-door' through which a buyer can begin their search for suppliers — they will, in effect, be well positioned as open market-places (see below). However, this vision does require that standard document type definitions (DTDs) can be adopted, at least across industry sectors.

2.2 Buyer-driven regime

Another class of trading environment is one in which an organisation has sufficient power as a buyer that they are able to dictate to their suppliers that in order to trade with the buyer they must participate in the buyer's own 'market-place'. This participation will extend to the publication of their product portfolio in a particular catalogue format and the acceptance of purchase orders via the market-place. This is a similar phenomenon to that experienced with the roll-out of EDI via the so-called ‘hub and spoke’ model — a major purchaser could dictate the terms under which they would do business with their many, but smaller, suppliers.

Goods purchased may be broadly grouped into two types:

- business-critical procurement, such as raw materials and components for a manufacturer or products for a retailer (direct goods),
- non-critical items used for maintenance repair and operations (MRO), such as stationery, IT equipment, etc — also known as indirect goods.

The requirements on systems to support these two types of purchase are slightly different. In the first case procurement is specified by a small group of specialised buyers, perhaps assisted by materials requirements planning (MRP) systems, who need to be able to source goods reliably, and quickly identify alternative suppliers in the event of a problem with the preferred supplier. In the retail sector, access to a broad supply base is particularly important. On the other hand, for indirect goods, purchases will be originated from personnel throughout the enterprise, and one of the major challenges is to rationalise the supplier base for particular products and drive down ‘rogue’ purchasing in order to maximise the ability of the enterprise to negotiate favourable trading terms with the supplier. The drive here is for the buying organisation to be able to significantly reduce the cost of their MRO expenditure, as well as to be able to accurately monitor and if necessary control expenditure of this kind. It is anticipated that the opportunity for providing MRO solutions will be greater than that for business-critical procurement.

Although the simplified view here is of a trading environment focused on serving one buyer organisation, there will be a very natural migration to open the environment up to other purchasing organisations who deal with the same supplier base. In this way a market-place that starts out as being conceptually driven by a single buyer will very quickly become an open market-place in the sense that it supports multiple suppliers and buyers.

2.3 Open regime

From the earlier discussion, it is apparent that suppliers adopting seller-driven catalogue applications and purchasers investing in buyer-driven (e.g. MRO) applications will naturally tend to become participants in open market-places in some form or other even if this is not a conscious decision. Moreover it will become increasingly important for suppliers to be able to publish their product data in a stand-alone catalogue as well as relevant open-markets.

Open markets may be private (supporting a closed buyer community) or public. The private markets will typically be characterised by services targeting business MRO procurement where a significant level of investment in an intranet procurement application, and the associated integration into back-office systems, will be necessary in order to participate as a buyer, while the public markets will probably emerge from community-of-interest Web sites and search engine portals.

Private open markets

There are a number of system developers such as Ariba, CommerceOne and Infobank developing an MRO procurement proposition to take to major corporations throughout the world. The proposition is that once a company has an intranet in place, it can be used to deploy an application to be accessed by anyone within the organisation who wishes to procure MRO type goods. The application can implement the processes used within the company to authorise requisitions, place the purchase orders and control the allocation of budgets. Bringing this information together means that rogue purchases can be brought under control, and the company can streamline its supplier base — increasing its ability to negotiate favourable trading terms. More automation of the procurement process is also an objective; for example, the Ministry of Defence [6] has identified that it has spent over US$ 120 to purchase a US$ 1.60 padlock, as a result of the paperwork and time spent on raising the purchase order.

The other major part of the proposition is that the system can be used to deliver an on-line product catalogue for each of the approved suppliers to the desktop of everyone in the organisation, so that employees have up-to-date product information available to them when required. The mechanism for keeping a purchaser’s intranet catalogue in step with a supplier’s main data source all vary slightly from one implementation to another (discussed below).

Similarly, there are also systems designed to support on-line market-places for direct-goods procurement. A good example of such a system is BT Trading Places [3].

Public open markets

Although buyers are readily able to find suppliers with well-known brands, it will still be a very laborious process for a buyer to investigate a wide range of potential suppliers without some kind of intermediary service working on the buyer’s behalf. In fact, although it has been a commonly held view that the Internet will lead to substantial disintermediation, there will be significant opportunities for new intermediary services perceived to be acting on a buyer’s behalf [7]. These new services will have the opportunity to become ‘portals’ through which buyers can access on-line suppliers. This trend is already starting with the major search engines offering shopping services, for example, see Excite Shopping [8]. Other players will specialise in specific vertical sectors.

Advances in technology, particularly the use of agents, will arguably have the greatest impact in this domain. Consumers deciding that they want to make a purchase will very often not be expert in that domain, and will therefore be receptive to assistance in deciding precisely what product to buy and where to buy it from.

Imagine someone wanting to buy a computer, but finding the whole subject quite bewildering. They may have a very good idea of what they want it for, e.g. to run an accounting package, word-processing and producing advertising brochures — but no idea of how this translates into what hardware and software to buy. To help such buyers towards their goal, an intermediary service could provide an advisor agent to help translate their need into a product specification. This agent role has been termed a ‘generic advisor’ [7], and a good illustration of this approach is PersonaLogic [9].

Secondly, the buyer will need to decide which manufacturer to approach and which model to buy. It is envisaged that a second type of agent, a ‘specific advisor’ [7] will be able to take the specification and provide the buyer with information from independent product reviews, electronic word of mouth and possibly the views of other similar purchasers to help make this decision. Excite Shopping [8] is now moving towards providing this kind of service for certain product areas.

Finally, the buyer needs to find the best supplier for the particular products identified as those needed to be purchased. There are a number of approaches being adopted by prospective open market services here to provide comparison shopping such as Anderson Consulting’s early BargainFinder agent or Jango as offered by Excite. An alternative approach is for the buyer to lodge a request-for-quotation (RFQ) with the service — the same approach as used in tendering for major contracts, but using the Web to scale the concept for the mass-market. In Fig 1 this approach corresponds to the bottom row labelled ‘requirements-led’ market. The efficiencies for the buyer come from having to define the requirement only once and not having to know about all of the prospective individual suppliers. The service will send this requirement (now a sales-lead) to suppliers who then have the opportunity to respond with a competitive offer. A good example of this approach is the Auto-By-Tel service [10] which specialises in providing a prospective car buyer with quotations from several dealers. This approach can result in lower costs to buyers, as a result of the suppliers having a lower cost of finding customers — they do not need to advertise as the sales-leads come to them from the intermediary service.

In the future it is envisaged that the distinction between RFQs and product finder agents will disappear from the user’s perspective — the intermediary service will be provided with a specification and then corresponding offers will be returned perhaps using both mechanisms to interface to the supplier. Also it is apparent that all of the discussion in this section has focused on the activity of sourcing the product rather than the mechanics of conducting the transaction. This is deliberate, because it is felt that the primary focus of open-markets will be in empowering non-expert purchasers to make very smart purchasing decisions.

3. XML

XML is set to be the enabler for the standardisation of interfaces between suppliers’ sites and buyers’ systems mentioned before. It is a syntax that will enable the standardisation of data structures up to and including complete blocks of data for a particular purpose, known as documents. Standardisation of the data structures, together with a standard transfer protocol such as HTTP, gives a standard interface. Although XML syntax itself has expressive power only a little greater than that of EDI, XML is set to find application in many areas other than trade data interchange. Also a document type definition and a style sheet can contain transformation and additional information in transportable form that needs to be embedded in a translator for EDI. This flexibility, portability and ubiquity will bring with it cheapness of the basic processing engine (to be included in future Web browsers, for instance) and a large skill base. This should ensure that all are able to benefit from it, whereas EDI has only really benefited the larger companies that could afford the translators and skilled personnel, and recoup costs through heavy usage.

4. Architectures and platforms to support open markets

In general, open markets will evolve from either buyer-driven or seller-driven markets. Open markets evolving from buyer-driven markets will do so to exploit an existing community of suppliers by recruiting new buyers. The evolution from seller-driven markets will be because buyers will find sourcing difficult if individual catalogues have to be interrogated separately.

A trading environment to support an open market can conform to a centralised, distributed, or hybrid model from the point of view of its system architecture.

4.1 Centralised

An environment designed to support a buyer-driven market is likely to be implemented as a centralised system. In evolving into an open market-place, a centralised system can typically go on to support a large number of different suppliers, many of whom may only have a small number of products to trade. Buying organisations are given simple electronic access to this range of suppliers, and suppliers will gravitate to the system if they know that the buyer is doing their business through this electronic system. It should also be simple and cost effective for small businesses to use. The BT Trading Places service based on the DeepSky (formerly GEMM) software is an example of a fully centralised system (see Fig 2). Suppliers enter product/ service information on to a central system and buyers search that same system.

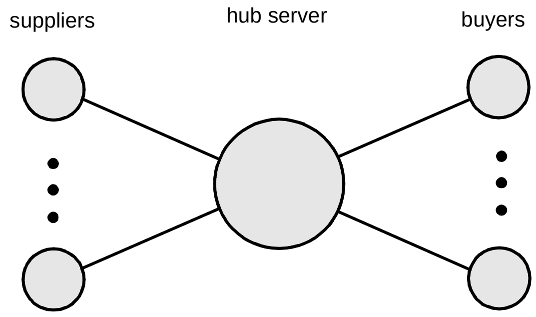

Fig 2 A centralised architecture.

The principal distinguishing characteristics of centralised systems are:

- centrally managed data schema,
- supports large numbers of suppliers efficiently on single system,
- needs to be replicated for security (with a single server, if that server fails, the whole service fails),
- service could be divided geographically or by product/ service category (but this would need further development of the current system),
- pure centralised service is not infinitely scalable, though it can grow large enough with current technology to cope with significant and usable portions of the commercial world, but eventually the database becomes too large and this limits the size — it is rather like trying to have a single telephone exchange for the whole world.

4.2 Distributed

A fully distributed system is arguably the correct one for the longer term, although it is not without its drawbacks.

Initially it can start by integrating existing fully featured catalogue systems, or at least suppliers' product data-sets. More suppliers are added by including further catalogue systems. It thus starts out life as a supplier-driven system. However, such an arrangement gives little sense of 'community', and gives scant help to buyers who have to point their browsers at a series of different Web catalogue sites. The answer is to provide the buying organisations with a buying system. Typical of this approach is the BT Supply Centre [3] service based on Infobank's 'Intrade^{TM}' software, which could be integrated into a fully distributed system (see Fig 3). Suppliers will enter product/service information on to their own catalogue. Buyers have their own software that takes a (buyer-) controlled index from all the suppliers' catalogues, permits local searching, and supports more detailed information retrieval from the remote catalogue systems, buyer controls, and ordering across the network of systems. Other examples include the Elcom system.

The principal distinguishing characteristics of fully distributed systems are:

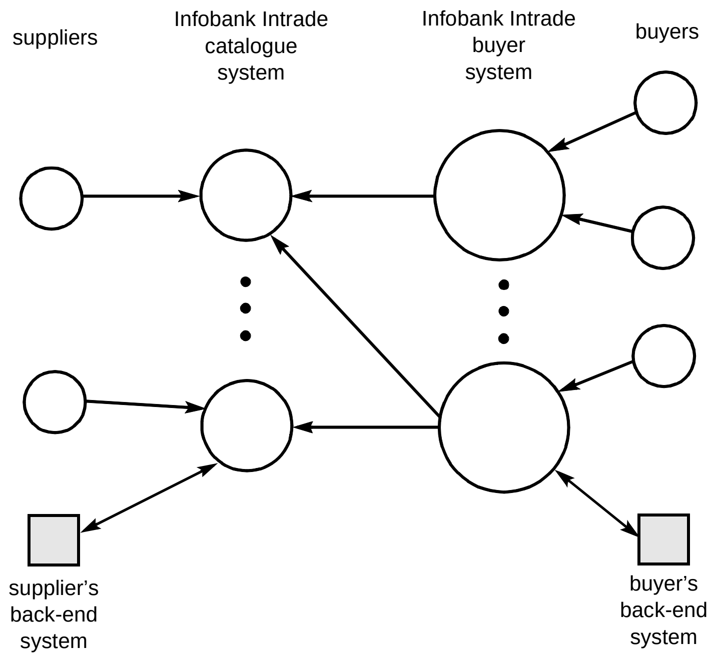

Fig 3 A distributed architecture.

- each supplier has its own data schema — buyers' systems either have fragments of each schema according to what they have decided to draw down, or each buying organisation has to manipulate the index into its own schema,
- each supplier has its own catalogue system,
- the service is potentially scalable (though buyers would need to connect to many catalogues — a potential management nightmare),
- although the overall system is logically distributed it does not have to be physically distributed — some, or all of the physical servers could be at the same physical site, which gives the advantages of enabling provision of common infrastructure (such as communications, routers and firewalls, housing, environment and system management),
- if a single server fails only one buying or supplier organisation is affected, whereas if servers are managed by a single entity (such as BT), '1 in *N* sparing' can be used to enhance service availability.

4.3 Hybrid

CommerceOne have adopted a hybrid approach. They collect all the suppliers data into a centralised system with a centrally managed schema, but have separate buyer software that can re-map the schema to one chosen by that buying organisation, and apply buyer controls (see Fig 4).

The principal distinguishing characteristics of this hybrid system are:

- centrally managed data schemata for suppliers — buyers can have their own categorisation schema if they (or another party) re-map it,
- supports large numbers of suppliers efficiently on central system,
- the central site needs to be replicated for security (with a single site, if that site fails, the whole service fails), whereas if it is a buyer system, it only affects that one buying organisation, not the whole service (could use '1 in *N* sparing' if the buyers' systems are managed by a single entity),
- the centralised part of the architecture poses a potential scaling problem — this either has to be accepted as a limitation on the growth of the service, or avoided by in fact following a distributed model within the apparently centralised part.

4.4 A vision for the future

While all the possibilities have some things to commend them, the best long-term basis would seem to be the distributed model, as this model will scale and is inherently reliable from an overall service point of view. An important component of this will be some kind of directory service, allowing buyers to readily locate discrete catalogues and market hubs as required. This model (see Fig 5) also lends itself to the incorporation of components supporting the other models, so long as all share common interfaces for the exchange of catalogue information, orders, etc. This common interface is the key to such an approach and needs to cover the data structures and their meaning, not just the

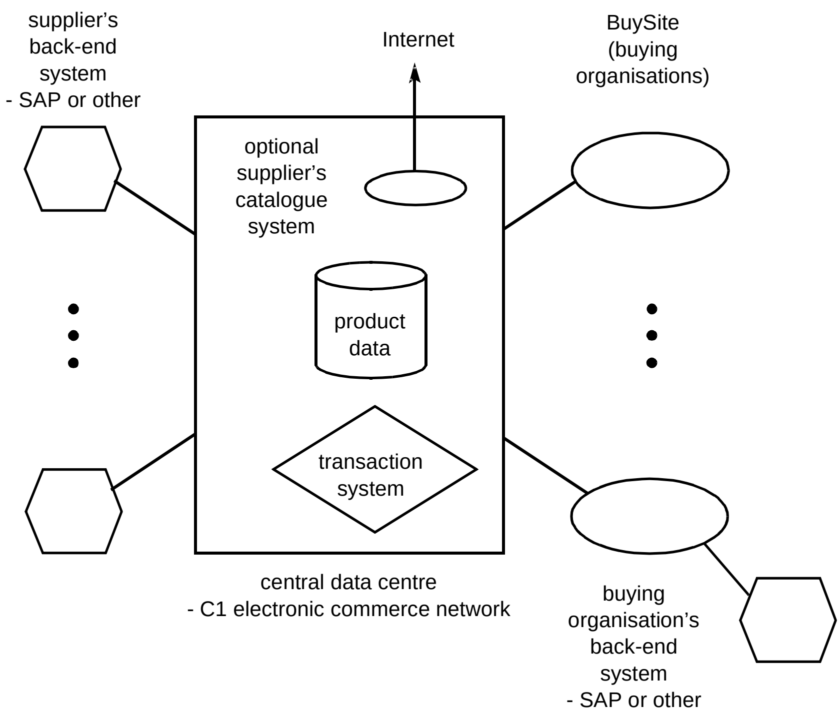

Fig 4 A hybrid architecture.

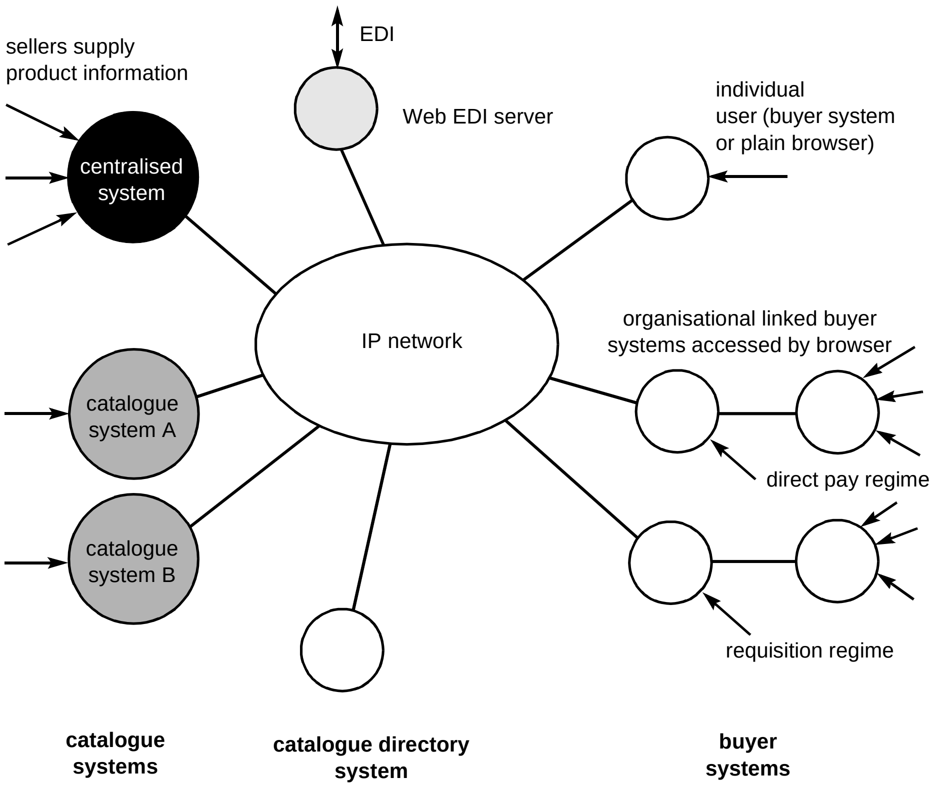

Fig 5 The distributed model for the future.

communications protocol. As mentioned above, XML is the favourite candidate for the syntax, but a great deal of work still needs to be done to specify the precise data structures and their meaning for this interface using the XML syntax mechanisms.

5. Other issues

Trading electronically does raise some issues that this paper does not cover in any detail. The two most obvious ones are how goods, once ordered, are delivered, and how they are paid for. A less obvious, but actually quite complex issue is how do I know that the entity I am trading with really are who they say they are? This issue of verification of identification and establishing an appropriate trust relationship are dealt with elsewhere [11, 12].

Delivery of goods divides into two. Physical goods have to be delivered physically. Charges for post or courier delivery can be a significant overhead charge on on-line shopping. Information, computer games and other programmes can be delivered digitally via the same channel as used for the trading (although bandwidth/delivery on-line time could be an issue — trading can be made a relatively low bandwidth activity, but the goods purchased may be large in terms of bytes).

For businesses the main payment mechanism will be paper invoices, or increasingly their electronic equivalent, although corporate credit cards and other such payment mechanisms can be used. For consumers, invoicing after the event of delivery is perhaps more open to fraud than when there has been face-to-face contact, although some companies may be prepared to take risks in the process of establishing trusted trading relationships. However, it is clear that payment before delivery is safer for the supplier, and there are various possibilities for achieving this. Payment mechanisms are described in more detail elsewhere [13].

Other issues that can arise in electronic trading include cross state and country selling, tax laws and payment, and legal issues concerned with data protection, sale of goods, consumer protection and so on.

A comprehensive trading environment will cater for all these issues. However, the current state of the art is that most applications only cover some of these elements electronically and rely on additional processes and systems for their completion.

6. Conclusions

On-line trading environments are still very much in their infancy, but are certain to become key to the way businesses, both large and small, trade in the future. It is already clear that there will be several different on-line trading models for different circumstances. Catalogue applications will continue to be important, but will only be viable as stand-alone market-places for suppliers who are confident that their brand will be strong enough to bring buyers to them. They will, however, be challenged by new open market-places that will build their own brand and offer buyers sufficient benefits to become the first point of call when sourcing a new product. On the other hand, the use of on-line trading for major buyers, particularly in the area of MRO will also flourish. Many suppliers may need to participate in the open Internet via a stand-alone catalogue, be accessible through a number of open-markets, and be able to sell to major corporates using MRO procurement systems. The challenge is for industry to implement standard interfaces for the eBusiness components to enable buyers and suppliers to interact through a variety of applications.

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